

10 Bài báo về AI agent , MCP , RAG

1. Agentic RAG Survey

- **Paper:** "Agentic Retrieval-Augmented Generation (Agentic RAG): A Survey On Agentic RAG"
- **GitHub:** <https://github.com/asinghesu/AgenticRAG-Survey>
- **PDF:** Có trên GitHub repository
- **Mô tả:** Khảo sát toàn diện về Agentic RAG, bao gồm reflection, planning, tool use và multi-agent collaboration

2. Adaptive-RAG

- **Paper:** "Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity"
- **GitHub:** <https://github.com/starsuzi/Adaptive-RAG>
- **PDF:** <https://arxiv.org/abs/2403.14403>
- **Mô tả:** Framework adaptive động lựa chọn chiến lược RAG phù hợp (no-retrieval, single-step, multi-step) dựa trên độ phức tạp của query, sử dụng classifier để predict complexity

3. Self-RAG

- **Paper:** "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection"
- **GitHub:** <https://github.com/AkariAsai/self-rag>
- **PDF:** <https://arxiv.org/abs/2310.11511>
- **Mô tả:** Framework cho phép LLM tự động retrieve, generate và critique thông qua self-reflection tokens, adaptive retrieval on-demand

4. PaperQA2 (Agentic RAG cho Scientific Papers)

- **Paper:** "High accuracy RAG for answering questions from scientific documents"
- **GitHub:** <https://github.com/Future-House/paper-qa>
- **PDF:** Có paper năm 2024 về superhuman performance (trong repo)
- **Mô tả:** Agentic RAG chuyên biệt cho tài liệu khoa học với citations

5. MCPToolBench++: Large-Scale MCP Tool Use Benchmark

- **Paper:** "MCPToolBench++: A Large Scale AI Agent Model Context Protocol MCP Tool Use Benchmark"

- **PDF:** <https://arxiv.org/pdf/2508.07575>
- **GitHub:** <https://github.com/mcp-tool-bench/MCPToolBenchPP>
- **Mô tả:** Benchmark quy mô lớn với hơn 4,000 MCP servers từ 40+ categories, đánh giá khả năng tool-use của LLM agents (Tháng 8/2025)

6. MCP Security and Maintainability Study

- **Paper:** "Model Context Protocol (MCP) at First Glance: Studying the Security and Maintainability of MCP Servers"
- **PDF:** <https://arxiv.org/pdf/2506.13538>
- **GitHub:** <https://github.com/modelcontextprotocol/servers>
- **Mô tả:** Nghiên cứu quy mô lớn đầu tiên về MCP servers, đánh giá bảo mật và maintainability

7. MCP Safety Audit

- **Paper:** "MCP Safety Audit: LLMs with the Model Context Protocol Allow Major Security Exploits"
- **PDF:** <https://arxiv.org/abs/2504.03767>
- **GitHub:** <https://github.com/johnhalloran321/mcpSafetyScanner>
- **Mô tả:** Công cụ audit bảo mật cho MCP servers, phát hiện vulnerabilities

8. MCP-Bench: Benchmarking Tool-Using LLM Agents

- **Paper:** "MCP-Bench: Benchmarking Tool-Using LLM Agents with Complex Real-World Tasks via MCP Servers"
- **PDF:** <https://arxiv.org/pdf/2508.20453>
- **GitHub:** <https://github.com/Accenture/mcp-bench>
- **Mô tả:** Benchmark với 28 live MCP servers, 250+ tools, đánh giá multi-step tasks phức tạp trên 20 LLMs (Tháng 8/2025)

9. RQ-RAG (Learning to Refine Queries)

- **Paper:** "RQ-RAG: Learning to Refine Queries for Retrieval Augmented Generation" (COLM 2024)
- **PDF:** <https://arxiv.org/abs/2404.00610>
- **GitHub:** <https://github.com/chanchimin/RQ-RAG>
- **Mô tả:** Học cách tinh chỉnh queries cho RAG

10. CRAG (Corrective RAG)

- **Paper:** "Corrective Retrieval Augmented Generation"
- **PDF:** <https://arxiv.org/abs/2401.15884>

- **GitHub:** <https://github.com/HuskyInSalt/CRAG>
- **Mô tả:** RAG với retrieval evaluator để cải thiện robustness, sử dụng web search khi retrieval không tốt